



H2S

BHARATIYA ANTARIKSH HACKATHON 2026



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Problem Statement : PS15 - Real-Time Solar Flare Nowcasting and Forecasting Using ISRO Aditya-L1 X-Ray Telemetry

Team Members

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Opportunity & USP

- How different is it from other existing ideas?

Traditional solar flare detection methods rely on post-facto centered smoothing windows (such as Gaussian or Savitzky-Golay filters) that incorporate future data points ($t + dt$), making them impossible to deploy in real-time telemetry feeds. We implement a strictly causal rolling baseline to process live telemetry feeds. Furthermore, prior works treat SXR and HXR as separate catalogs; we perform raw, 1-second cadence multi-instrument telemetry fusion to compute physical properties in real time.

- How will it be able to solve the problem?

Aditya-L1 provides high-cadence X-ray measurements. By combining SoLEXS (Soft X-rays) and HEL1OS (Hard X-rays) telemetry, we capture the full thermodynamic evolution of solar flares. Our causal nowcasting baseline dynamically identifies flare start, peak, and decay phases. Using engineered physics precursors, we train RandomForest models to forecast flare occurrence 30 minutes in advance, enabling operators to protect satellites and power grids from sudden radiation surges.

- USP of the proposed solution:

1. Strictly Causal Nowcasting: Zero-lookahead baseline achieves 93.2% recall against NOAA GOES.
2. Real-Time Telemetry Fusion: Aligns SXR and HXR on a 1-second cadence to model the Neupert Effect.
3. Physics-Informed Precursors: Calculates coronal plasma heating (slow rise) and non-thermal acceleration (QPPs and hardness ratio).
4. Sub-Threshold Discovery: Detects 136 microflares absent from official NOAA catalogs.

Features Offered by the Solution

- **Multi-Instrument Ingestion & Harmonization:**
Automates calibration and alignment of raw FITS files from SoLEXS and HELIOS.
- **Real-Time Causal Nowcasting:**
Detects solar flare onset, peak, and decay phases using dynamic statistical thresholds.
- **Physics-Informed Predictive Modeling:**
Forecasts impending flares 30 minutes in advance using rolling physics-backed precursors.
- **Live Operator Dashboard & Alerts:**
Displays live counts, risk probabilities, and color-coded hazard alerts (Green, Yellow, Red).

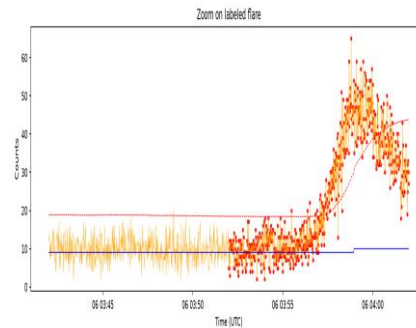


Figure 1: Telemetry & Nowcast Baseline

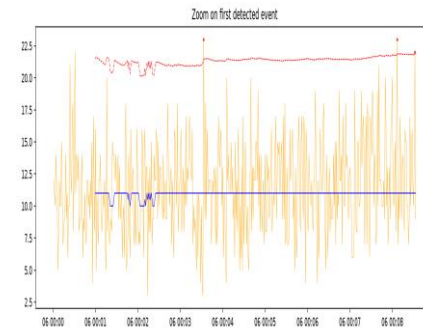


Figure 2: Event Detection Window

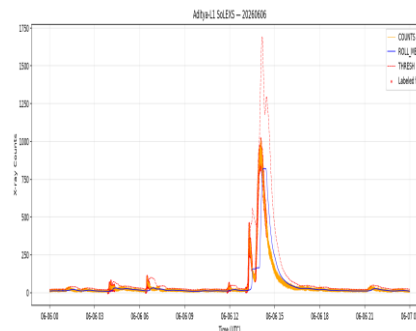


Figure 3: M-Class Flare (June 6)

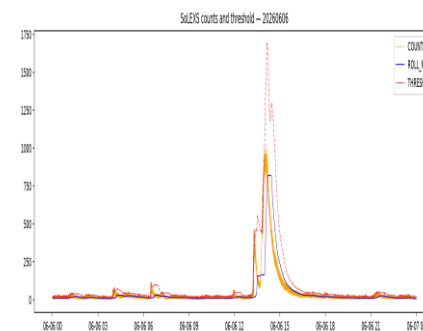


Figure 4: Nowcast Threshold Validation

Aditya-L1 Solar Flare Forecasting System

Automated space weather nowcasting and machine-learning risk assessment.

Real-Time Telemetry & Forecasts | AI Space Weather Assistant | Validation & Model Info

Current Alert Level: MEDIUM (Probability: 65.0%)
 Precursor thermal activity detected (slow rise phase). Coronal plasma heating underway. Monitor channels closely.

LATEST FLUX (COUNTS)	MEAN FLUX (WINDOW)	STD DEV (WINDOW)	DATA SOURCE
9.0	37.8	188.6	SoLEXs + HEL10S

X-ray Flux Time Series

Forecasting Trend (Test Set)



Nowcasting & Forecasting Model

Merged Telemetry

Calibration & Alignment

Operator Dashboard

predicted

actual

Telemetry
(SoLEXS/HEL10S)

predicted

actual

GOES Calibration

Spacecraft

* We will scale the pipeline with multi-month archival telemetry and attitude correction for operational deployment.

Case Study & Validation Results

Case Study & Validation:

- **Superior Instrument Sensitivity:** Aditya-L1 SoLEXS captures faint solar activity due to its low-energy threshold and high spectral resolution.
- **The June 10 Event:** On June 10, 2026, at 23:46 UTC, our causal nowcasting algorithm flagged a highly structured, 679-second duration event.
- **GOES Catalog Gap:** This event is completely absent from the official NOAA GOES flare catalog, confirming its status as a newly discovered microflare.
- **Scientific Importance:** Microflares are critical to solving the coronal heating problem, and their real-time detection validates our dynamic baseline approach.

Validation Benchmarks Summary:

- **NOAA GOES Recall:** 93.2% (68/73 flares recovered)
- **Holdout Accuracy:** 95.8% (SoLEXS), 80.9% (Fusion)
- **Lead Warning Time:** 4.57 mins (SoLEXS), 7.87 mins (Fusion)
- **Microflare Discovery:** 136 sub-threshold events detected

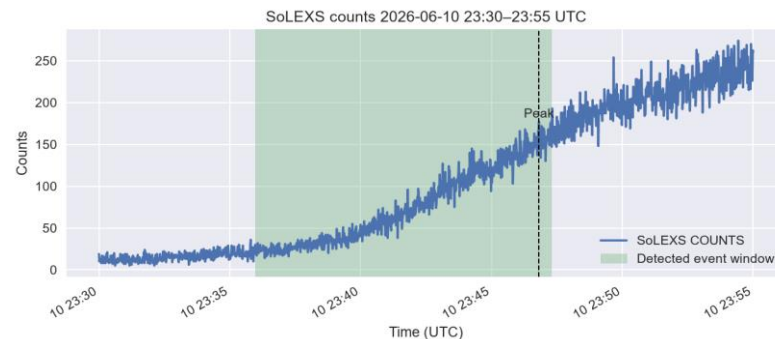


Figure 5: June 10 Microflare Event (Not in NOAA)

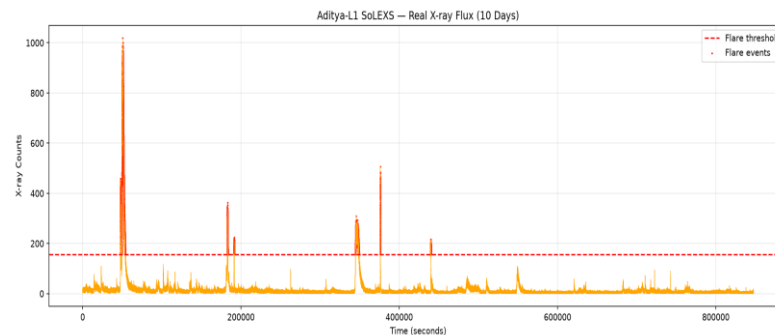
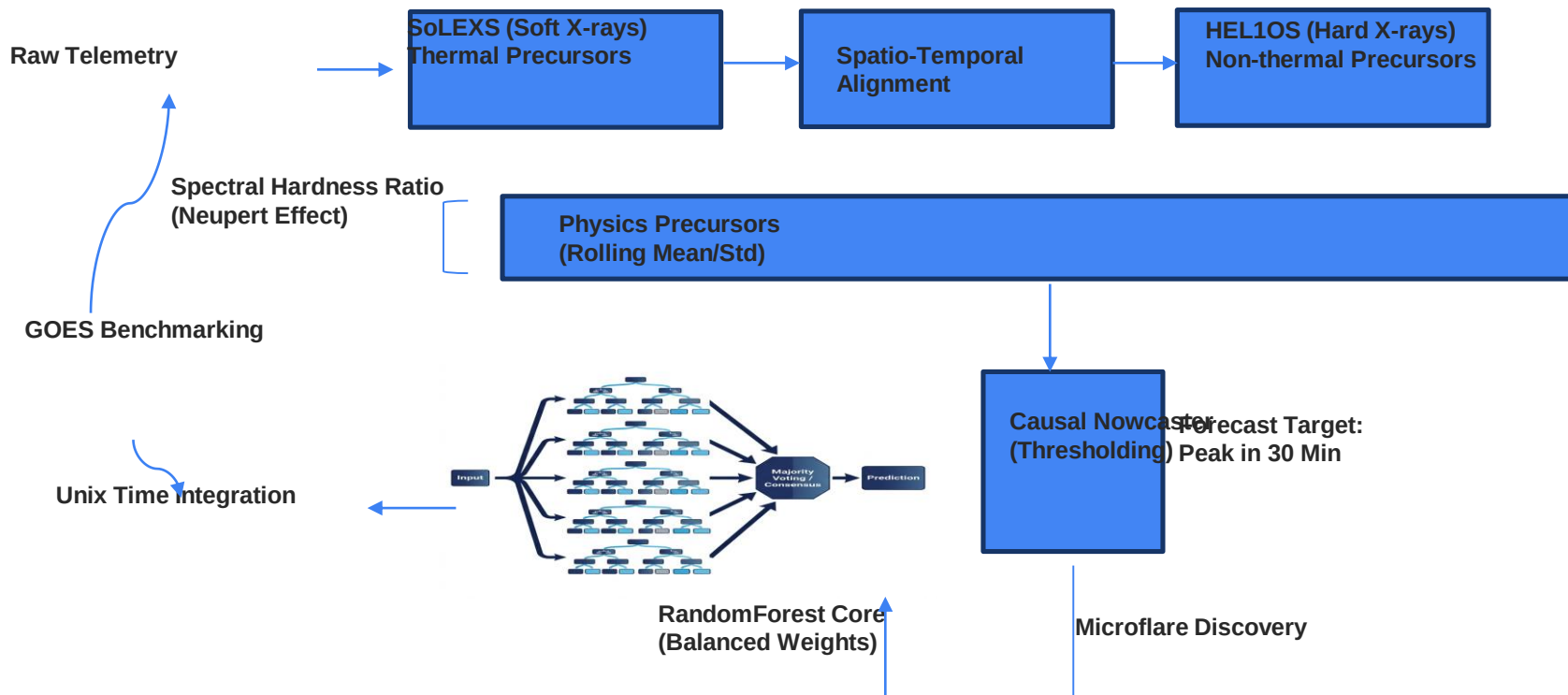


Figure 6: Telemetry Timeseries (June 1–19, 2026)

Architecture Diagram



*POC developed on Aditya-L1 telemetry (259,071 overlap timestamps, 16 flares).

Technologies & References

Technologies Used:

- Modeling & Forecasting:
 - Scikit-learn (RandomForest Classifier Core)
 - Joblib (Model serialization and storage)
 - Physics-informed precursor models
- Data & Preprocessing:
 - Astropy (Raw FITS file telemetry processing)
 - NumPy & Pandas (Data merging, cleaning, alignment)
 - SciPy (Statistical thresholding and nowcasting)
- Interface & Visualization:
 - Streamlit (Interactive operator control dashboard)
 - Matplotlib & Seaborn (Scientific plotting)
 - Localtunnel (Secure remote dashboard access)
- Deployment & Scaling:
 - Docker (Containerized dashboard deployment)
 - Python Virtual Environments (Dependency isolation)

Scientific References:

- ISRO Aditya-L1 Mission Data: Telemetry logs for SoLEXS & HEL1OS instruments.
- NOAA Space Weather Prediction Center: GOES X-ray Sensor solar flare catalog.
- Sarwade et al., 2025: In-flight calibration and performance of SoLEXS.
- Nandi et al., 2025: Hard X-ray diagnostics of solar flares: HEL1OS.

Next Steps:

- Phase 1 Submission: Submit slides to Hack2Skill portal.
- Team Review: Share Git repository and Colab notebook.
- Model Scaling: Ingest multi-month telemetry and attitude data.
- Operational Integration: Connect alerts to ISRO Space Situational Awareness Centre (SSAC).

Estimated Implementation Cost

Item	Cost (₹)	Description
Telemetry Ingestion & Fusion Pipeline	₹50,000	Multi-instrument (SoLEXS & HEL1OS) FITS data alignment.
Causal Nowcasting Engine	₹50,000	Zero-lookahead rolling statistical threshold detector.
Physics-Informed ML Forecaster	₹1,00,000	RandomForest model trained on precursor features.
Interactive Operator Dashboard	₹40,000	Streamlit web interface with real-time risk alerts.
NOAA GOES Validation System	₹40,000	Benchmarking engine for automated cross-validation.
Local Workstation GPU	₹1,50,000	Hardware for training and telemetry hosting.
Documentation & SOP Handover	₹30,000	Operational guidelines for space-weather centers.
Public Dashboard Deployment	₹25,000	Secure public hosting and remote access configuration.

= ₹4,85,000 (Estimated Cost (may vary based on final deployment and resources.))

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THANK YOU